

Noam Rothschild

Israel
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Security Researcher @ bugsec

Security researcher passionate about systems programming, operating systems, virtualization, networking, and reverse engineering. I enjoy building low-level software from scratch to understand how computers work.

Personal Projects: <https://github.com/NoamRothschild>

Personal Website: <https://noamrothschild.github.io/projects>

EXPERIENCE

BUGSEC, Rishon LeZion — Security Researcher

July 2026 - PRESENT

Penetration testing a physical device on a classified project. Assisting with **embedded device** security assessments involving **firmware extraction**, hardware interfaces, and **network attack surface analysis**.

PROJECTS

Hypervisor & Operating System — Zig, Assembly

Started writing a **type 1 hypervisor**. It currently sets up the VMCS, the EPT for each guest, **vm entry and exit** routines, and intercepts `hlt` and `cpuid` requests. The hypervisor is built on top of one of my kernels:

Implemented significant components of an experimental operating system, including **higher-half kernels**, GDT/IDT setup, IRQ handlers, **interrupt handling**, **paging**, and a **physical memory allocator**.

Operating System: <https://github.com/NoamRothschild/Mini-OS>

Hypervisor: <https://github.com/NoamRothschild/hypervisor>

Distributed MMO Game Server — Python, Redis, TCP/UDP

Designed and led a five-person team in development of a **distributed MMO backend** across multiple physical servers with **Redis** messaging, **encrypted TCP/UDP networking**, persistence, inventory, combat, and **dynamic load balancing**.

GitHub: <https://github.com/NoamRothschild/cyber-project>

LANGUAGES & TOOLS

Python · C · C++ · x86/x64
Assembly · Zig · JavaScript · Java

Linux · Git · Redis · GDB · QEMU
· Docker · Bash · Make · Vim

Daily driving arch linux

TECHNICAL INTERESTS

Operating Systems

Hypervisors

Embedded Programming &
Security

Distributed Systems

AWARDS

1st Place Computer Science Final
Projects Competition:

Won 1st place among 60~
students for “Discord in
Assembly”, an x86 assembly
implementation of a Discord
client.

Winner Magshimim Regional
Vault Competition:

Won the year’s regional cyber
competition of the Magshimim
program in the Darom region.

Assembly Standard Library & Networking Stack — x86 Assembly (NASM)

Built a **libc-free standard library** in x86 assembly featuring **sockets**, **shared memory**, file I/O, **HTTP/WebSocket** support, **SHA-1**, Base64, and Discord/Voxel Space clients.

GitHub: <https://github.com/NoamRothschild/asm>

Interpreter & Compiler — Zig, Assembly

Developed an **interpreter** for the 8086, supporting most of its instruction set, including **labels**, **branching**, and an **interactive REPL**. Designed as both a standalone and an embeddable Zig dependency, used to **automate grading** of assembly assignments in my school.

Also developed a **compiler prototype** featuring **Pratt parsing** and **assembly code generation** for math expressions.

8086 interpreter: <https://github.com/NoamRothschild/asm-interpreter>

Pratt Parser: <https://github.com/NoamRothschild/pratt-parser-zig>

Magshimim Cyber Projects — C++, Java

Developed multiple networking and backend projects as part of the Magshimim cybersecurity program, including a **multiplayer chess backend**, a **distributed trivia server** (Kahoot clone), and a Java client.

GitHub: <https://github.com/NoamRothschild/MagshiTrivia>

EDUCATION

Atid School For Science Lod — Computer Science & Cyber

Sep 2024 - PRESENT

10 Units in Computer Science; 5 units in Software Engineering + 5 Units in Cyber. 5 Units in Math, Physics, and English.

Magshimim, Kiryat Gat — Cyber

Sep 2024 - PRESENT

Coursework from Magshimim and School includes:

- Computer Networks
- Computer Architecture
- Systems Design
- x86 Assembly
- Reverse Engineering
- Orchestration of Large Scale Systems (System Design)

